B15 Embeddings and clustering

SentenceTransformers, HDBSCAN and cluster validation.

B15-L1 Sentence embeddings for open-ended answers

One model, recorded and cached. Compare with cosine similarity, and guard against zero vectors.

```python
from sentence_transformers import SentenceTransformer

model = SentenceTransformer("sentence-transformers/all-MiniLM-L6-v2")
vectors = model.encode(answers, batch_size=64, normalize_embeddings=True)
```

Encode with sentence-transformers	SentenceTransformer(name).encode(texts, normalize_embeddings=True). Not run in this course's build.
Cosine similarity for unit vectors	The dot product. Otherwise divide by both lengths.
Mixing embedding models	Never. Each model defines its own space.
Cache key for a vector	Text ID, text hash, model name and version.

B15-L2 HDBSCAN, clusters with a noise label

Let density decide the clusters, keep noise as its own answer, and summarise counts, noise share and balance before naming themes.

```python
from sklearn.cluster import HDBSCAN

labels = HDBSCAN(min_cluster_size=15, copy=True).fit_predict(vectors)
```

HDBSCAN noise label	-1: points in sparse regions. Report the share, never treat it as a cluster.
HDBSCAN main setting	min_cluster_size (default 5). min_samples makes it more conservative.
HDBSCAN in scikit-learn 1.8	sklearn.cluster.HDBSCAN. Pass copy=True: the default changes in 1.10.
Read a clustering	Noise share, number and size of clusters, then samples from each.

B15-L3 Validate clusters before you name them

Check separation without noise, stability across reruns, and meaning by reading, before any cluster gets a name.

Three checks before naming clusters	Separation (silhouette), stability (ARI across reruns), meaning (read and name independently).
Silhouette and HDBSCAN noise	Exclude -1 first. silhouette_score treats it as a cluster (0.70 vs 0.555 in this lesson's example).
Adjusted Rand index	1 for identical groupings, about 0 for chance, can be negative, ignores label names.
Examples on a theme slide	Random verbatims, not hand-picked ones.